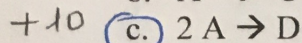
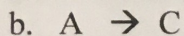
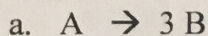


1. (25 points) Three gas phase reactions have the same reaction rate dependence to the concentration of the reactant.



Which of these reactions will exhibit the highest conversion in a PFR? Explain.

2 molecules of A make 1 molecule of D $\rightarrow$ mix decreases

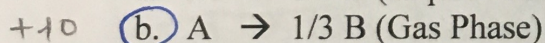
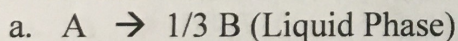
+15

$\rightarrow C_A$ increase $\rightarrow X_A$ increase.

2. (25 points) Why does the conversion for a gas-phase reaction in a PBR depend on the pressure?

Because the pressure drop is significant in PBR for gas-phase reaction.

3. (25 points) Two reactions (one gas phase and one liquid phase) have the same reaction rate dependence to the concentration of the reactant.



Which of these reactions will exhibit the highest conversion in a PFR? Explain.

+15 In liquid phase, pressure drop is insignificant.

In gas phase, due to pressure drop $\rightarrow C_A \uparrow \rightarrow X_A \uparrow$

4. (25 points) A reaction rate equation is given by $r_A' = k' P_A$. What are the units of k' ?

$$[r_A'] = \frac{[\text{mol}]}{[\text{kg}][\text{s}]} = k' \cdot [\text{atm}]$$

$$\Rightarrow k' = \frac{[\text{mol}]}{[\text{kg}][\text{s}][\text{atm}]}$$

-5 miss 1 term

+10 not relevant